VIRAJ KUMAR VISHWAKARMA

Full-Stack Engineer • AI / ML Engineer • Competitive Programmer

+91 6264449774 | virajvishwakarma672@gmail.com | linkedin.com/in/viraj-kumar-vishwakarma | github.com/virajkvk18 | Jabalpur, MP, India

PROFESSIONAL SUMMARY

Computer Science undergraduate (AI & ML specialization) with end-to-end experience building full-stack web applications, multi-agent LLM systems, AI training environments, and deployed SaaS products. Proficient in React.js, Next.js, FastAPI, LangChain, OpenRouter, and Docker. Solid foundation in Data Structures & Algorithms with 190+ problems solved on LeetCode and Codeforces. Experienced in integrating NLP/ML models, designing RESTful APIs, building recommendation engines, and deploying containerized applications with CI/CD pipelines. Passionate about shipping products at the intersection of applied AI and scalable software engineering.

EDUCATION

B.Tech — Computer Science Engineering (AI & ML Specialization) *2023 – 2027 (Expected)*

Baderia Global Institute of Engineering and Management | Jabalpur, Madhya Pradesh

- Core coursework: Artificial Intelligence, Machine Learning, Algorithm Design, DBMS, Operating Systems, and Modern Web Architectures.
- Active competitor in national-level hackathons, AI agent challenges, and competitive programming contests.

Senior Secondary (PCM) *Graduated 2023*

St. Joseph Senior Secondary School | Jabalpur, Madhya Pradesh

TECHNICAL SKILLS

Languages: Python, C++, JavaScript (ES6+), TypeScript, SQL, HTML5, CSS3

Frontend: React.js, Next.js (SSR/SSG), Tailwind CSS, Shadcn/UI, Chrome Extension APIs, Responsive & Accessible Design (WCAG)

Backend & APIs: FastAPI, Node.js, Express.js, REST API Design, WebSockets, JWT, OAuth2, Stripe API, Docker, CI/CD

AI / ML: LangChain, OpenRouter, Langfuse, OpenAI API, HuggingFace Transformers, BERT, Scikit-learn, TensorFlow, PyTorch, RAG Pipelines, Vector Embeddings (FAISS), Multi-Agent Systems, OpenEnv, Prompt Engineering, A/B Testing

Databases: PostgreSQL, MongoDB, Redis, DynamoDB basics

Data: Pandas, NumPy, Matplotlib, Seaborn, ETL Pipeline Design, Anomaly Detection

Tools: Git, GitHub, Docker, Vercel, VS Code, Postman, Linux, YAML/TOML

CS Fundamentals: Data Structures & Algorithms, OOP, DBMS, OS, System Design, Concurrency Basics

TECHNICAL PROJECTS

Smart Interview Assistant · Next.js · FastAPI · OpenAI API · Whisper · PostgreSQL *Mar 2026*

- Built a full-stack AI web app that accepts a resume + job description and auto-generates tailored, role-specific interview questions using GPT-4 with structured prompt engineering.
- Integrated OpenAI Whisper for speech-to-text transcription of candidate answers; NLP scoring layer evaluates keyword density, confidence markers, and answer completeness, returning structured feedback in under 2 seconds.
- Designed a FastAPI backend with PostgreSQL persistence for session history and score tracking; React/Next.js frontend with real-time streaming feedback display and a candidate-friendly dashboard.
- Achieved 94% transcription accuracy on benchmark audio samples; system handles 50+ concurrent sessions via async FastAPI workers and Redis-backed queue.

TaskForge — Multi-Tenant SaaS Project Management Platform · Next.js · Node.js · PostgreSQL · Docker · Stripe · CI/CD *Feb 2026*

- Architected a full-stack SaaS project management tool (Jira/Notion-inspired) with multi-organization support, JWT + OAuth2 authentication, and role-based access control (RBAC) covering Admin, Manager, and Member roles.
- Integrated Stripe API for subscription billing with webhook-driven plan enforcement; supports Free, Pro, and Enterprise tiers with usage limits enforced server-side.

- Containerized with Docker Compose; configured GitHub Actions CI/CD pipeline for automated testing and zero-downtime deployment to Vercel (frontend) and Railway (backend).
- Designed for 1,000+ concurrent users with indexed PostgreSQL queries, connection pooling (PgBouncer), and optimistic UI updates reducing perceived latency by 60%.

RecSys — Personalized Recommendation Engine · Python · Scikit-learn · TensorFlow · FastAPI · Redis · PostgreSQL *Jan 2026*

- Engineered a hybrid recommendation system combining collaborative filtering (matrix factorization via SVD) and content-based filtering using TF-IDF item embeddings, trained on a 100K+ user-interaction dataset.
- Deployed trained model as a low-latency FastAPI inference endpoint; Redis caching layer serves precomputed top-N recommendations with sub-50 ms response time for repeat users.
- Implemented A/B testing framework to compare collaborative vs hybrid models; hybrid approach achieved 87% precision@10 and a 23% improvement in click-through rate over baseline.
- Built an async user-event ingestion pipeline (PostgreSQL) that continuously retrains the model on a 24-hour rolling window, keeping recommendations fresh without manual intervention.

SynthAI — AI Productivity Chrome Extension · JavaScript · Chrome Extension APIs · OpenAI API · HuggingFace *Jan 2026*

- Developed a Manifest V3 Chrome Extension that injects a floating AI assistant panel into any webpage; features include one-click page summarization, key-insight extraction, and highlighted-text Q&A.
- Integrates OpenAI GPT-4o for summarization and a HuggingFace NER model for entity extraction; summarizes a 5,000-word article in under 3 seconds with a structured bullet-point output.
- Built a background service worker for session state and a popup UI for site-time tracking and reading history; all data stored locally using Chrome Storage API — zero server cost, full privacy.
- Published to Chrome Web Store; works across 10,000+ websites with adaptive content extraction handling dynamic (React/Vue) and static pages via DOM MutationObserver.

IncidentMind — AI Agent Training Environment (OpenEnv) · Python · FastAPI · OpenEnv YAML · Docker · LangChain *Mar 2026*

- Engineered a spec-compliant OpenEnv environment simulating real-world SRE incident response workflows for AI agent training; designed tool-gated action space (log analysis, service restart, alert routing) with a structured JSON observation schema.
- Implemented multi-objective reward function balancing MTTR minimization and false-positive penalties; FastAPI server exposes /reset, /step, /observe endpoints packaged in Docker for reproducible evaluation.
- Built LangChain-based agent inference pipeline with stdout-formatted action logging fully compatible with competition evaluation harnesses.

CitizenPulse — Multi-Agent Health Classification Pipeline · Python · LangChain · OpenRouter · Langfuse · FastAPI *Feb 2026*

- Designed a two-agent LangChain pipeline for citizen health data classification in the Reply AI Challenge 2026 (MirrorLife sandbox): Agent 1 performs intake normalization and triage scoring; Agent 2 executes LLM-powered classification via OpenRouter.
- Integrated Langfuse for full observability — tracing token usage, latency per agent step, and end-to-end classification accuracy across 500+ test records.
- Emitted JSON-validated output payloads consumed directly by the MirrorLife simulation API, with zero schema validation errors across all submission runs.

RAGify — Retrieval-Augmented Generation Knowledge Assistant · Python · LangChain · FAISS · OpenAI Embeddings · Next.js *Dec 2025*

- Built an end-to-end RAG pipeline enabling natural language Q&A over custom document corpora: ingests PDFs and text files, chunks with overlap, embeds via OpenAI text-embedding-3-small, and indexes in FAISS for semantic retrieval.
- Next.js frontend with real-time streaming chat interface displays source citations alongside answers; source attribution achieved 91% factual grounding on domain-specific document sets, significantly reducing hallucination.
- Handles corpora of 500+ documents with sub-300 ms retrieval latency; supports multi-document context windows using LangChain's MapReduce summarization chain.

DevMesh — Real-Time Collaborative Code Editor · Next.js · TypeScript · WebSockets · Monaco Editor · Node.js *Dec 2025*

- Architected a browser-based collaborative coding platform with real-time multi-user editing powered by a Node.js WebSocket server; implements operational transform-based diff synchronization to handle concurrent edits with zero data loss.
- Integrated Monaco Editor (VS Code core) for syntax highlighting across 20+ languages; cursor broadcasting and presence indicators update in under 80 ms across all connected clients.
- Deployed on Vercel with edge-optimized Next.js API routes; session replay implemented via an append-only event log enabling full edit history reconstruction.

COMPETITIVE PROGRAMMING

- LeetCode: 132+ problems solved — Contest Rating 1416. Focus on dynamic programming, graph algorithms, binary search, and space-time complexity optimization.
- Codeforces: 60+ problems solved; consistent participation in rated global rounds — strong in greedy, constructive algorithms, and number theory.
- HackerRank: Problem Solving (Basic) Certification — verified proficiency in core algorithmic logic and data structures.
- C++ DSA Proficiency: deep implementation experience in trees, graphs, segment trees, tries, DP optimizations, and advanced sorting.

CERTIFICATIONS & ACHIEVEMENTS

- Reply AI Challenge 2026 — Built CitizenPulse multi-agent health classification system for the MirrorLife sandbox competition.
- Meta Scalar Hackathon 2026 — Developed and submitted MedTriage, an AI-powered medical triage API with full containerized deployment pipeline.
- HackerRank Problem Solving (Basic) Certification.
- Self-directed coursework: LangChain, RAG systems, Docker, OpenEnv agent frameworks, HuggingFace Transformers, and A/B testing methodologies.

EXTRACURRICULAR ACTIVITIES & INTERESTS

- Open-Source & Research: Proactive self-learner exploring agentic AI, LLM observability, distributed systems, and emerging web technologies through hands-on builds.
- Fitness & Sports: Committed to structured training and competitive sports — cultivates discipline, mental resilience, and pressure management directly applicable to engineering.
- Learning Philosophy: 'Build to learn' approach — consistently ships side projects and hackathon submissions to internalize new frameworks beyond academic curriculum.